Day 1 — Introduction to Data Analytics

Understanding Data, Analytics, the Data Analyst Role and the Analytics Mindset

DAY 1 • 1.5 HOURS

Learning Objectives

LEARNING OUTCOMES

- Define Data Analytics and explain why organizations use it.
- Understand the role and responsibilities of a Data Analyst.
- Recognize how data supports evidence-based business decisions.
- Understand the high-level flow from data to insight to decision.
- Develop an analytical mindset: ask the right question before choosing a tool.

What is Data Analytics?

CORE CONCEPT

- Data Analytics is the process of examining data to discover useful information, patterns and insights.
- The objective is not simply to produce numbers or charts — it is to support meaningful decisions.
- A strong analytics process connects: Business Question → Data → Analysis → Insight → Decision.
- Analytics can be applied across sales, finance, operations, marketing, HR, healthcare, education and engineering.

From Data to Decision

The Analytical Chain

1. Identify the problem
2. Define the question
3. Collect relevant data
4. Prepare and clean the data
5. Analyze and visualize
6. Communicate insight
7. Support a decision

Key Question

1. What decision are we trying to improve?
2. What evidence would help us make that decision?
3. Which data is relevant?
4. How will we know whether the analysis is useful?

Why Data Analytics Matters

INDUSTRY PERSPECTIVE

- Organizations generate large volumes of operational and customer data.
- Analytics converts raw observations into information that can be interpreted.
- Analytical evidence can help identify trends, compare alternatives, detect unusual behavior and measure performance.
- The value of analytics depends on the quality of the question, data, method and interpretation.

Examples of Analytics Questions

Business / Industry

Why did sales change this quarter?

Which products or regions contribute most to revenue?

Which customers are becoming inactive?

Where are operational delays occurring?

Engineering / Research

How does rainfall relate to runoff?

Which factors explain variation in demand?

Are there unusual observations in the data?

Can historical data improve prediction?

The Role of a Data Analyst

CAREER CONTEXT

1. Understand the business or research problem.
2. Identify and obtain relevant data.
3. Clean, organize and validate datasets.
4. Perform descriptive and exploratory analysis.
5. Create meaningful visualizations and reports.
6. Interpret results in context.
7. Communicate findings clearly to technical and non-technical stakeholders.

Data Analyst: Technical + Analytical + Communication Skills

SKILL FRAMEWORK

- Technical: Excel, SQL, Python, statistics and visualization tools.
- Analytical: problem formulation, critical thinking, pattern recognition and interpretation.
- Communication: dashboards, reports, presentations and concise explanation of findings.
- Professional practice: reproducibility, documentation, data quality and responsible use of data.

Data Analytics vs Data Science vs Machine Learning

CONCEPTUAL MAP

Aspect	Data Analytics	Data Science	Machine Learning
Primary Focus	Understanding data, patterns, performance and insights	Broader discipline involving analytics, statistical modelling, experimentation and predictive methods	Using algorithms to learn patterns from data
Typical Objective	Understand what happened and why	Understand, model and predict what may happen	Make predictions, classifications or other data-driven outputs
Key Activities	Data analysis, EDA, reporting, visualization, dashboards	Analytics, statistical modelling, experimentation, predictive analysis	Model training, prediction, classification and pattern learning
Common Tools	Excel, SQL, Power BI, Tableau	Python, SQL, Statistics, BI, ML tools	Python, Scikit-learn, TensorFlow, PyTorch
Example Question	“Why did sales decrease?”	“What factors influence customer churn?”	“Which customers are likely to churn?”
Role in This Program	Foundation	Developed through Statistics, Python, SQL and BI	Advanced application
Learning Progression	Analytics	Statistics → Python → SQL → BI → Predictive Methods	Machine Learning

The Analytics Mindset

PERSPECTIVE

- Start with the question, not the software.
- Ask: What does the variable represent? How was the data collected?
- Check data quality before interpreting patterns.
- Separate correlation from causation.
- Quantify uncertainty where appropriate.
- Translate technical output into a decision-relevant conclusion.

Mini Case Study — Retail Sales

Business Case: Investigating a Decline in Monthly Sales

From Data to Insights for Better Decisions

1 Scenario

A retailer observes a decline in monthly sales.



"Sales have been decreasing over the last few months."

2 Key Question

What should an analyst investigate before concluding why sales declined?



3 Possible Data

Relevant data to analyze:

- **Date** (time period)
- **Region** (location)
- **Product** (category / item)
- **Quantity** (units sold)
- **Price** (selling price)
- **Discount** (offers, promotions)
- **Customer segment** (age group, type)
- **Channel** (online, store, etc.)

4 Analysis Path

Validate data
Check for missing values, outliers or data issues

↓

Summarize sales
Total sales, growth rate, key metrics

↓

Compare periods
Month-over-month, year-over-year

↓

Examine segments
By region, product, customer segment and channel

↓

Visualize patterns
Use charts and dashboards

↓

Formulate insights
Evidence-based conclusions into recommendations

5 Example Visualizations

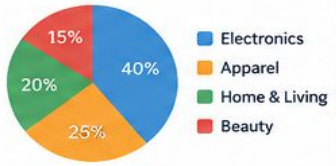
Monthly Sales Trend



Sales by Region



Sales by Product Category



Sales by Channel



6 Expected Outcome

Evidence-based insights to identify the **key reasons for the sales decline** and **data-driven recommendations** to **improve performance.**

"Data turns questions into answers, and insights into action."

Day 1 Lab — Dataset Exploration

HANDS-ON

Open a small business dataset in Excel.

Identify rows, columns, observations and variables.

Classify variables as categorical or numerical where appropriate.

Check for missing values, duplicates and inconsistent entries.

Write three analytical questions that could be answered using the dataset.

Share one question and explain why it matters.

Day 1 Activity — Think Like an Analyst

ACTIVE LEARNING

- Individual task: Choose a familiar domain such as sales, education, finance, healthcare, transport or engineering.
- Write one business/research problem.
- Convert the problem into 2–3 measurable analytical questions.
- List the data required to answer those questions.
- Identify one possible limitation or data-quality concern.
- Class discussion: Compare how different questions require different data.

Day 1 Recap

TAKEAWAYS

Data Analytics connects data with decisions.

The Data Analyst works across problem definition, data preparation, analysis, visualization and communication.

Good analysis begins with a clear question and reliable data.

Tools such as Excel, SQL and Python are means to perform analysis — not substitutes for analytical thinking.

Next: Data Analytics Lifecycle and the stages of an analytics project.

Exit Check — 5 Questions

ASSESSMENT

1. What is the primary purpose of Data Analytics?
2. Why should an analyst define the question before selecting a tool?
3. Name three responsibilities of a Data Analyst.
4. What is the difference between raw data and an analytical insight?
5. Give one real-world analytics question from your domain.

Use Case 1 — Netflix: Recommendation Analytics

80%+

Real-world evidence of analytics in action

Netflix reported that more than 80% of shows were discovered through recommendations in a 2017 company analysis. The system uses signals such as viewing history, ratings, similar-member behavior, title information, language, device and viewing duration to personalize what appears to users.

Source: Netflix, 2017; Netflix Help Center (recommendation system).

Use Case 2 — UPS: Route Optimization

130M+ miles

Real-world evidence of analytics in action

UPS reported in 2021 that Dynamic ORION and earlier ORION deployments together were associated with more than 130 million miles of driving reduction and about 10 million gallons of fuel saved annually. This is an example of descriptive data becoming optimization and operational decision-making.

Source: UPS Investor & Analyst Day, 2021.

Use Case 3 — Amazon India: Inventory Analytics

35% fewer stockouts

Real-world evidence of analytics in action

Amazon India reported in September 2026 that sellers following AI-driven inventory recommendations experienced 35% fewer stockouts. The example shows how analytics can move from historical sales data toward recommendations that influence inventory decisions.

Source: Amazon India Press Center, September 2026.

Use Case 4 — Netflix: Scale of Business Data

97B+ hours

Real-world evidence of analytics in action

Netflix reported that members watched more than 97 billion hours globally in the first half of 2026. At this scale, analytics supports measurement of engagement, content discovery, audience patterns and business decisions across genres and languages.

Source: Netflix, What We Watched: First Half of 2026.

What the Numbers Teach Us

1

Measure

Define the business outcome before choosing a tool.

2

Diagnose

Use data to understand what is happening and why.

3

Predict

Estimate what may happen next using statistical/ML models.

4

Act

Convert an insight into a business decision or workflow.

Takeaway: analytics is not “doing charts”; it is a disciplined path from evidence to action.

Day 1 Mini-Case — From Business Question to Metric

Business Question

Why are monthly sales falling?

Data Needed

Sales, product, region, customer, price, date

Analysis

Trend → segment → root-cause analysis

Decision

Pricing, inventory, promotion or targeting

Class discussion: Which metric would you inspect first — revenue, units sold, average order value, conversion rate, or customer retention?

2026 Job Market — What Employers Are Asking For

4,016

public Data Analyst job postings identified in India over the 89 days ending 17 Sep 2026; ~293/week.

SQL • Python • Power BI

These were the three most requested skills in the analyzed India postings.

25%

of the analyzed postings offered either remote or hybrid work.

AI + Data

WEF identifies big-data specialists and AI/ML specialists among the fastest-growing job categories toward 2030.

Sources: India job-posting analysis (Jun 20–Sep 17, 2026); World Economic Forum Future of Jobs Report 2025.

Expected Salary — Data Analytics Roles in India (2026)

₹6.35 LPA

Indeed's reported average base salary for Data Analysts in India, updated 5 Sep 2026.

₹5–10 LPA

Glassdoor's current base-pay range shown for Data Analysts in India; average base pay about ₹6 LPA.

₹5.15 LPA

Indeed reported average for entry-level Data Analysts; senior Data Analyst average was about ₹9.72 LPA.

Important: salary varies substantially by city, company, experience, domain, interview performance and skill depth.
These are market indicators, not guaranteed starting salaries.

The 2026 Entry-Level Skill Stack

1. Excel

Cleaning, formulas, Pivot Tables, reporting

2. SQL

SELECT, JOIN, GROUP BY, CTEs, window functions

3. Power BI / Tableau

Dashboards, KPIs, data modelling, storytelling

4. Python

Pandas, NumPy, visualization, automation

5. Statistics

Descriptive statistics, probability, hypothesis testing

6. AI Literacy

Prompting, validation, GenAI-assisted analysis

Students should demonstrate the full workflow — raw data → SQL/Python → BI dashboard → business recommendation.

Career Path — Where Can a Data Analyst Grow?

0–2 years

Junior / Data Analyst

₹4–8 LPA

Reports • Data cleaning • SQL • Dashboards

2–4 years

Data / BI Analyst

₹7–14 LPA

KPIs • Automation • Stakeholder analysis

4–7 years

Senior Analyst / Analytics Consultant

₹12–22 LPA

Lead analysis • Mentoring • Complex problems

6–10+ years

Analytics Manager

₹20–35+ LPA

People leadership • Strategy • Transformation

Indicative India 2026 market ranges • Actual CTC varies by company, city, domain, skills and experience

Salary Progression — Indicative India Ranges

Junior / Data Analyst	₹4–8 LPA
Data / BI Analyst	₹7–14 LPA
Senior Analyst / Analytics Consultant	₹12–22 LPA
Analytics Manager	₹20–35+ LPA
Product Analyst	₹10–22+ LPA
Data Scientist	₹12–30+ LPA

₹6.35 LPA

Indeed's reported average base salary for Data Analysts in India (Sep 2026).

Use salary ranges as market indicators — not guaranteed offers.

Indicative India 2026 market ranges • Actual CTC varies by company, city, domain, skills and experience

What Drives Salary Growth?

Technical Depth

SQL • Python • Power BI/Tableau • Statistics • Data modelling

Business Impact

Revenue • Cost reduction • Productivity • Customer outcomes

Domain Knowledge

Finance • Healthcare • Retail • BFSI • SaaS • Operations

Communication

Storytelling • Stakeholder management • Executive presentations

AI & Automation

GenAI • Prompting • Workflow automation • ML-assisted analytics

Portfolio Evidence

Real datasets • GitHub • Dashboards • End-to-end case studies

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How This 4-Month Course Maps to the Career Path

Excel	Reporting fundamentals	Junior Analyst
SQL	Data extraction + business queries	Data / BI Analyst
Statistics	Evidence-based reasoning	Senior Analyst
Python	Analysis + automation	Senior Analyst / Data Science
Power BI	Dashboards + data modelling	BI Analyst / Analytics
Tableau	Advanced visualization	BI / Product Analytics
Power Automate	Workflow automation	Analytics Automation
Prompt Engineering	AI-assisted analytics	AI-enabled Analyst
Machine Learning	Prediction + segmentation	Data Science / Advanced Analytics

Indicative India 2026 market ranges • Actual CTC varies by company, city, domain, skills and experience

What Should a Student Be Able to Demonstrate?

✓ **Can I write SQL?**	JOINs • GROUP BY • CTEs • Window functions
✓ **Can I analyse data?**	Cleaning • EDA • Statistics • Business interpretation
✓ **Can I build a dashboard?**	KPIs • Data model • DAX/calculations • Storytelling
✓ **Can I automate?**	Python • Power Automate • Reproducible workflows
✓ **Can I use AI responsibly?**	Prompt • Validate • Debug • Check hallucinations
✓ **Can I explain the result?**	Insight → recommendation → measurable business impact

Certificates open doors; demonstrated problem-solving skills help you progress.

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Salary & Market Data — Sources and Interpretation

Indeed India	Data Analyst salary pages, accessed Sep 2026	Average base salary indicators for India Data Analyst roles.
Glassdoor India	Data Analyst salary estimates, 2026	Reported base-pay range and employee-submitted compensation data.
U.S. Bureau of Labor Statistics	Occupational Outlook / Data Scientists	Useful global benchmark for the broader analytics/data science pathway.
World Economic Forum	Future of Jobs Report 2025	Global employer survey on fast-growing roles and skills toward 2030.

Salary ranges in this presentation are instructional market bands, not official pay scales or guaranteed compensation.

Why Analytics + AI Is a Long-Term Skill

+33.5%

U.S. Bureau of Labor Statistics projects data-scientist employment growth from 2024 to 2034.

170M

new jobs projected globally by 2030 in WEF's Future of Jobs 2025 analysis, alongside 92M displaced jobs.

86%

of surveyed employers expected AI and information-processing technologies to transform their business by 2030.

Interpretation for students: learn tools, but build transferable abilities — problem framing, statistics, data reasoning, communication and responsible AI use.

What Should a Student Show in a Job Interview?

Project 1

Excel + Power BI dashboard

Project 2

SQL analytics case

Project 3

Python analysis

Project 4

AI-assisted analytics

Project 5

End-to-end capstone

Message: “Do not show only certificates. Show evidence that you can solve a business problem with data.”